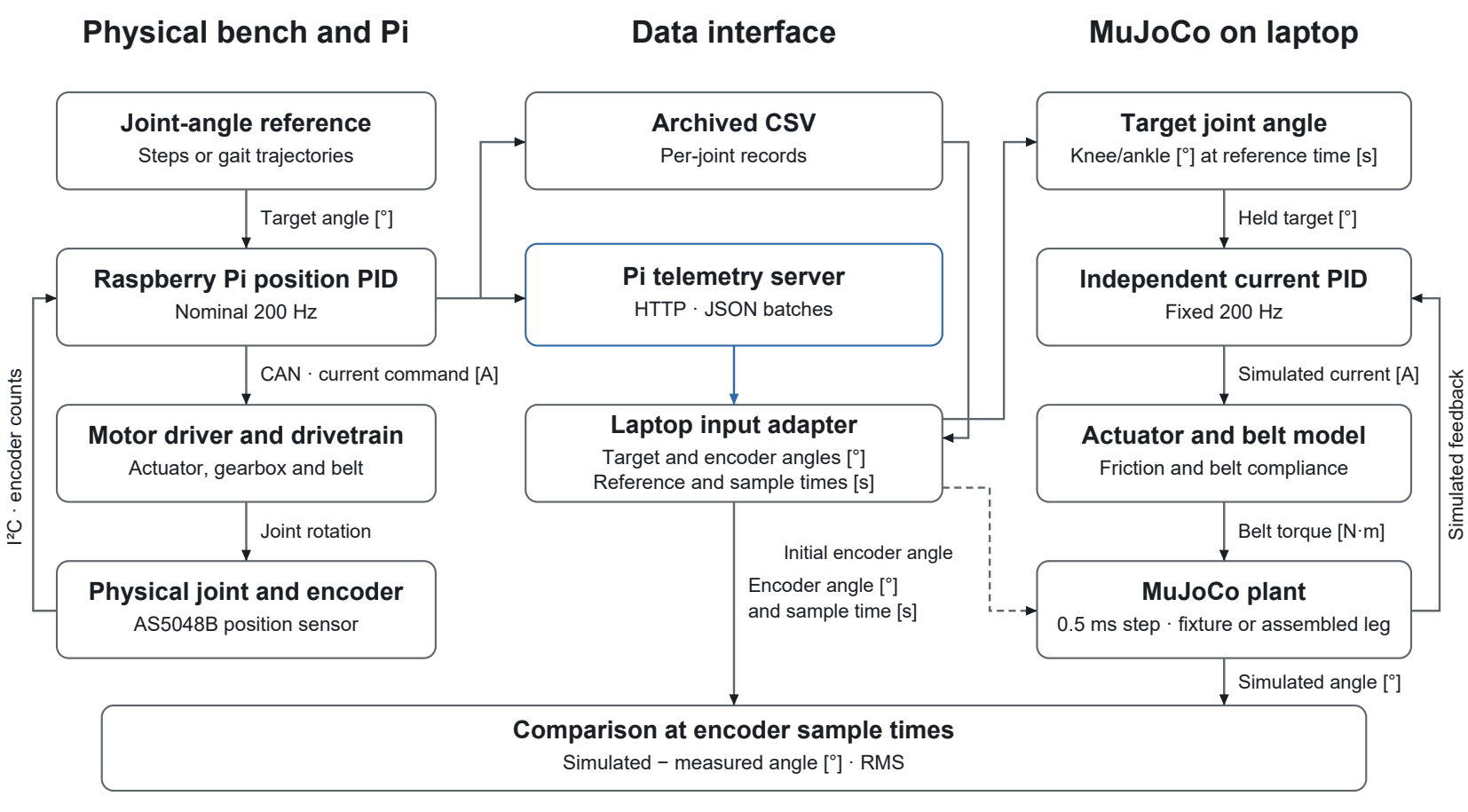

Physical bench and Pi
Data interface
MuJoCo on laptop
Joint-angle reference
Steps or gait trajectories
Target angle [°]
Raspberry Pi position PID
Nominal 200 Hz
CAN · current command [A]
Motor driver and drivetrain
Actuator, gearbox and belt
Joint rotation
Physical joint and encoder
AS5048B position sensor
I²C · encoder counts
Archived CSV
Per-joint records
Pi telemetry server
HTTP · JSON batches
Laptop input adapter
Target and encoder angles [°]
Reference and sample times [s]
Initial encoder angle
Encoder angle [°]
and sample time [s]
Target joint angle
Knee/ankle [°] at reference time [s]
Held target [°]
Independent current PID
Fixed 200 Hz
Simulated current [A]
Actuator and belt model
Friction and belt compliance
Belt torque [N·m]
MuJoCo plant
0.5 ms step · fixture or assembled leg
Simulated angle [°]
Simulated feedback
Comparison at encoder sample times
Simulated − measured angle [°] · RMS